# A SMOOTH RATIONALLY CONNECTED QUINTIC HYPERSURFACE IN $\mathbb{P}^9_K$ OVER A $\mathrm{cd}(K) \leq 1$ FIELD WITHOUT RATIONAL POINTS

ARGUS OPEN RESEARCH PROJECT

Abstract. Combining Ax's 1970 counterexample to Artin's conjecture ($\mathrm{cd}(K) \leq 1$, $\mathrm{char}(K) = 0$) with an exact resultant smoothness check and Campana/Kollár–Miyaoka–Mori Fano rational connectedness, we record an explicit smooth rationally connected degree-five hypersurface $X \subset \mathbb{P}^9_K$ with $X(K) = \emptyset$.

## 1. Main Proposition

**Theorem 1.1.** *Let $K$ be the characteristic-zero field of cohomological dimension $\mathrm{cd}(K) \leq 1$ constructed by Ax (1970, Theorem 1), and let $\Phi(X_0, \ldots, X_9)$ be the associated homogeneous quintic form in $10$ variables over $K$ whose binary quintic building block is $f(z, w) = (z^2 - sw^2)(z^3 + sw^3)$. Then:*

(1) *The resultant of $f(z, 1) = (z^2 - s)(z^3 + s)$ and its $z$-derivative is*

$$\mathrm{Res}_z(f, f_z) \;=\; -108\, s^7 (s-1)^2 \;\neq\; 0$$

*for the parameter $s \notin \{0, 1\}$, so $f$ is separable and the diagonal-block quintic hypersurface $X = V(\Phi) \subset \mathbb{P}^9_K$ is smooth over $K$.*

(2) *By adjunction, $\omega_X \cong \mathcal{O}_X(5 - 10) = \mathcal{O}_X(-5)$ is anti-ample, so $X$ is a smooth Fano hypersurface of dimension $8$ in characteristic zero and is therefore rationally connected.*

(3) *By Ax's theorem, $\Phi$ has no nontrivial zero in $K^{10}$, so $X(K) = \emptyset$.*

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